Adaptive Energy-Bounding Approach for Robustly Stable Interaction Control of Impedance-Controlled Industrial Robot With Uncertain Environments

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***Abstract*—Even though impedance control approaches are useful for robot interaction control, they can guarantee stability only for the assumed range of the environments. This paper presents a new adaptive energy-bounding approach (EBA) that can maintain a desired contact force at steady state while guaranteeing robust contact stability for impedance-controlled industrial robots that are contacting with very uncertain environments beyond the assumed range of the environments. The proposed adaptive EBA that is applied to impedance-controlled industrial robots interacting with incompletely specified real environments is an extension of the haptic EBA, which was previously developed for stable haptic interaction with virtual environments. Moreover, the adaptive EBA estimates online control parameters to improve performance while assuring stability for the system. Theoretical analyses on the performance and experimental results demonstrate the effectiveness of the proposed approach.**

***Index Terms*—Haptics, impedance control, industrial robot, passivity, robust stability.**

I. Introduction

INDUSTRIAL robots are widely used in many factory automation areas. The robot applications including contact tasks need not only motion control but also force control to perform interaction works safely between the robot and environment. During the past 30 years, many force control methods have been proposed [1]–[4] to accomplish contact tasks stably. Among them, the impedance control [5] and hybrid position/force control [6] are two major strategies in the robot force control.

Commercial industrial robots usually have an accurate and robust position controller. Taking advantage of these features, the position-based impedance control (PBIC) strategies [7], [8] were proposed to perform interaction tasks with the position-controlled industrial robots. The PBIC, then, does not require any modification of the conventional position controller and allows natural contact transition process from free space to contact interaction without any switching algorithms. However, the PBIC may not guarantee contact stability when the environmental parameters are changed drastically beyond the ranges that were assumed in the design phase. For example, a humanoid robot may step on the unknown ground or a wearable robot may encounter with large change of human contact impedance while lifting a heavy object. The model-based robust control methods for impedance-controlled robots (e.g., [7], [9]), therefore, may require redesign of the impedance parameters when environmental parameters are changed significantly because the stability is closely related to the environmental parameters.

To prevent potentially unstable problem caused by variable environmental parameters, many adaptive impedance control strategies have been proposed together with offline estimation of the environmental parameters [10] or with online identification of environmental properties for unknown or very uncertain environments. For example, adaptive impedance controls for teleoperation were developed based on estimates of the remote environmental dynamics [11], based on the online estimation of operator's arm damping [12], or based on real time identification for the Hunt–Crossley dynamic models of the environments [13]. On the other hand, intelligent force controls based on neural network [14] and fuzzy [15] algorithms may also be combined with the basic methods to complement their drawbacks and to increase performances. However, most advanced algorithms are usually difficult to implement in practice.

Previously, we proposed the haptic energy-bounding algorithm (*haptic EBA*) [16] for robustly stable haptic interaction with any linear, nonlinear, delayed, passive, and active virtual environments. By this robust stable characteristic of the haptic EBA, we think that the haptic EBA concept can also be applied to the interaction control with real environments that may be very uncertain. Thus, this paper presents a new adaptive energy-bounding algorithm (*adaptive EBA*) that uses the position-controlled industrial robot to achieve a desired contact force while guaranteeing robust contact stability, *especially when the environmental stiffness is drastically changed*. Even though we have adopted the previous concept of the haptic EBA, the application of the haptic EBA to the position-controlled industrial robot requires reformulation of the passivity-based control theory as well as new interpretation of the main control parameters. In particular, the energy behavior of the environment

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needs to be included in the theory development for adaptation to uncertain real environments, which had not been considered in the previous haptic EBA. This resulted in a modified Lemma 2 that is different from the original Lemma 2 which is a proof of the passivity condition related to uncertain environments in the haptic EBA [16]. Furthermore, full performance analyses of the proposed adaptive EBA are newly performed to explain analytically how and why we can get excellent and robust performance. Along with the new formulation, we explained more clearly how to determine control parameters too.

The proposed scheme needs to change first the position-controlled mode of the industrial robot to a PBIC mode because the haptic EBA was developed for the impedance-type haptic devices such as the PHANToM and the Omni [17]. The PBIC is then designed against the environmental parameter ranges that had been determined by the experiments and experiences. Note that since these environmental parameters may be uncertain or variable because of limited experiments and experiences, the adaptive EBA estimates online control parameters to improve the performance.

The proposed algorithm has a relatively simple implementation on top of the PBIC of the industrial robot. Moreover, the proposed adaptive EBA preserves the original impedance control performance if the environmental parameters are within the original ranges. To the best of the author's knowledge, this is the first attempt to apply the EBA to position-controlled real industrial robot that has high inertia and large gear ratio.

The remainder of this paper is organized as follows: the following Section II briefly discusses the PBIC of the industrial robot. Section III presents the theory development for the proposed adaptive EBA on top of the PBIC for robust contact stability. Detailed discussions are also presented on how to set some control parameters that can be tuned online with the theoretical explanations. In addition, analytical performance analyses are presented to show effectiveness of the proposed approach. Next, Section IV describes experimental setup and presents typical experimental results showing effectiveness of the proposed approach. Final section summarizes conclusions and future works.

II. Review of PBIC

For the position-controlled industrial robots performing constrained motion tasks, impedance control strategies may take advantage of the robustness and accuracy of the position controller of the well-established industrial robots. Unlike conventional controls that are designed to track given reference input signals, impedance control regulates the mechanical impedance of the manipulator. This means that impedance control can change robot's original impedance to a target impedance to reduce very high contact impedance of the position-controlled robot, which, therefore, can provide stable interaction with environments.

For the commercially available off-the-shelf robots that usually have closed architectures, some researchers have developed the improved PBIC [7], [18]. Fig. 1 shows a generic form of the PBIC architecture, where X_0 is the nominal position, X_e is the environment location, G_p is the position-controlled robot transfer function that has an internal position feedback loop, and

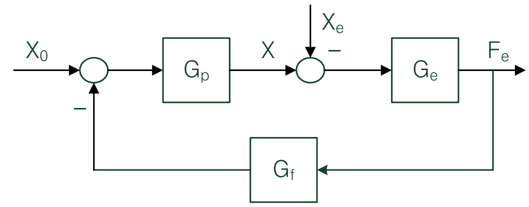


Fig. 1. PBIC block diagram for the industrial robot.

G_e block is the transfer function representing an environment. In Fig. 1, the transfer function G_f is the improved impedance controller that can be designed such that

$$G_f(s) = \tilde{G}_p^{-1}(s)\, G_t^{-1}(s) = \frac{s^2 + as + b}{b} \frac{1}{M_t s^2 + B_t s + K_t} \tag{1}$$

where $\tilde{G}_p$ is a minimum phase estimate of the position-controlled industrial robot transfer function G_p and where G_t is the target impedance function that can be modeled as a simple second-order system with target mass (M_t), damper (B_t), and stiffness (K_t) elements. Note that the PBIC for contact tasks with an industrial robot has big advantages: a simple architecture as compared with other algorithms and no modification of conventional position controller of the robot [9].

For the robust contact stability, the PBIC used the following condition for single-input single-output (SISO) linearized interaction system [9] as

$$\xi_t \geq \frac{1}{2}\left(\sqrt{1 + 2\kappa} - 1\right) \tag{2}$$

where ξ_t is the target damping ratio $(\xi_t = B_t/2\sqrt{M_t K_t})$ and $\kappa = K_e/K_t \gg 1$ is the stiffness ratio between the environmental stiffness (K_e) and the target stiffness (K_t). The system, however, may become unstable when the actual environmental stiffness exceeds the value that was assumed in the design phase. Moreover, the robust control may be too conservative if the assumed value range K_e in (2) is set to a large value for wider stiffness ranges. For instance, an overdamped system designed for robust stability usually shows slow response and sluggish behavior.

III. Adaptive EBA With PBIC

The haptic EBA [16] that provides robust stability for various virtual environments can also be applied to the interaction control with real environments that may be very uncertain. The haptic EBA in [16], however, was formulated with constant parameters for satisfying passivity for any virtual environments. No adaptation strategy, however, was tried for performance improvement. The proposed adaptive EBA, therefore, is reformulated theoretically with redefined parameters that can be online adjusted for uncertain real environments. The haptic EBA was developed for the impedance-type haptic devices such as the Omni and the PHANToM [17] that are torque controlled, lightweight, and back-drivable. In order to apply the haptic EBA to the position-controlled industrial robot, a PBIC needs to be designed first, which transforms the position-controlled robot to the force-controlled robot.

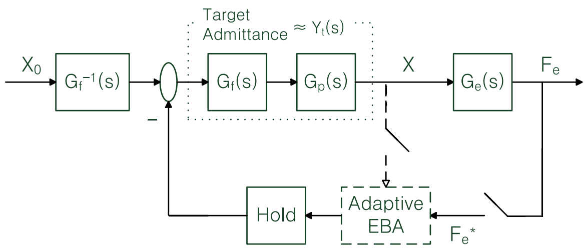
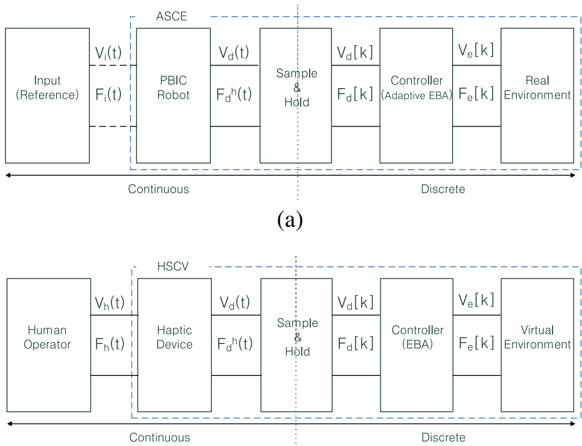


Fig. 2. Sampled-data system of the converted PBIC with EBA.



(a)

(b)

Fig. 3. Network models of PBIC and typical haptic system. (a) Impedance-controlled industrial robot. (b) Haptic system.

A. *Network Model Representation*

Since the theory of the haptic EBA was derived from the network model with a passivity condition, a network model for the PBIC industrial robot in Fig. 1 is needed. Assuming X_e is zero and G_p is a linearized function that has a physical zero-order-hold (ZOH*)* for simplicity of analysis, a block diagram manipulation changes Fig. 1 to Fig. 2 by moving the impedance controller (G_f) in the feedback path and by adding the outer-loop sampling effect and the proposed adaptive EBA. In other words, the impedance-controlled robot system is considered as a continuous plant that is to be controlled. The final network model in Fig. 3(a) can then be directly derived from Fig. 2. Note that the proposed adaptive EBA in Fig. 2 regulates the discrete contact force F_e^* that corresponds to the virtual environmental force $F_e\,[k]$ in Fig. 3(b). Note in Fig. 3(a) that the velocity signal V_i represents a virtual conjugate variable for defining a port energy [19].

Fig. 3 shows both network models of the PBIC industrial robot system and the typical impedance-type haptic system. Key differences between the impedance-controlled robot system and the haptic system are: 1) input subsystem (reference signal versus human operator), 2) device subsystem (impedance-controlled industrial robot versus impedance-type haptic device), and 3) environment type (real versus virtual).

B. *Passivity Condition for Stable Interaction of PBIC Industrial Robot*

A passivity condition to the PBIC industrial robot can then be derived straightforwardly from Fig. 3(a). Control input in the first network in Fig. 3(a) can be assumed passive because $G_f^{-1}\,(s)$ is composed of passive elements. Thus, the whole system will be passive if the one-port network in Fig. 3(a) is passive by the following definition [20]:

Definition: The one-port network system is passive if the energy absorbed by the network over any period of time $[0, t]$ is greater than or equal to the increase in the energy stored in the network over the same period; i.e.,

$$\int_0^t F_i\,(\tau)\,v_i\,(\tau)\,d\tau \geq V_{\mathrm{ASCE}}\,(t) - V_{\mathrm{ASCE}}\,(0)\,, \quad \text{for } \forall t \geq 0 \tag{3}$$

where $V(\cdot)$ represents the energy stored in the network.

Note that the one-port is composed of the combined *ASCE* system, where *A, S, C, and E* stand for the Admittance model of the impedance-controlled industrial robot that is obtained from $G_f\,(s)G_p\,(s)$, the Sample and hold, the Controller (adaptive EBA), and the Environment, respectively.

In order to derive a stable interaction control algorithm, the energy of the combined *ASCE* subsystem is divided into two subsystems: the admittance subsystem $(1/Z_t^{-1})$ and the *SCE* subsystem. First, the net energy flow into the two-port admittance system over $0 \leq t < nT$ is derived as

$$E_A\,[n] = \int_0^{nT} \left\{F_i\,(t)\,v_i\,(t) - F_d^h\,(t)\,v_d\,(t)\right\} dt \tag{4}$$

where T is the outer-loop sampling period (which is in the order of 50 Hz in this paper) and discrete value $E_A[n]$ is equal to continuous value $E_A(nT)$ for nonnegative integer n. Second, the net energy flow into the combined one-port *SCE* subsystem over $0 \leq t < nT$ can be derived as

$$\begin{aligned} E_{\mathrm{SCE}}\,[n] &= \int_0^{nT} F_d^h\,(t)\,v_d\,(t)\,dt \\ &= \sum_{k=0}^{n-1} F_d[k]\,(x_d\,[k+1] - x_d\,[k])\,. \end{aligned} \tag{5}$$

Note in (5) that the constant force F_d^h through the ZOH can be put outside of the integral and exact integration of the velocity *v(t)* generates position difference between the future position $x_d[k+1]$ and the current position $x_d[k]$ [16]. The *SCE* subsystem may become active even though the *CE* subsystem is passive because of the nonzero phase lag in the sample and hold. Therefore, the sample and hold must be fully included in the passivity condition unless very high frequency sampling (usually 30 times faster than the system bandwidth for being treated as a continuous system) is used, which is usually not the case for many industrial robots.

From (4) and (5), the whole robot-environment interaction system becomes then passive if

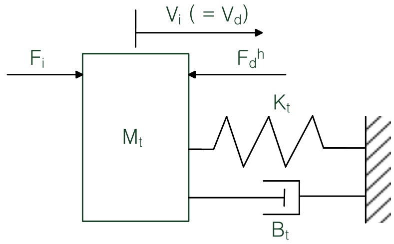


Fig. 4. Single-DOF model for improved impedance-controlled industrial robot.

$$E_A[n] + \sum_{k=0}^{n-1} F_d[k]\,\Delta x_d[k+1] \geq V_{\mathrm{ASCE}}[n] - V_{\mathrm{ASCE}}[0] \quad (6)$$

where $(x_d[k+1] - x_d[k])$ is represented by $\Delta x_d[k+1]$ for simplicity.

C. Adaptive EBA for PBIC Industrial Robot

Since the sample and hold operator and the *CE* subsystem may contain several energy generation factors, the second term $(\sum_{k=0}^{n-1} F_d[k]\,\Delta x_d[k+1])$ in (6) can be negative. One strategy to ensuring passivity is that the generated energy must be dissipated properly by the energy dissipation capability in the admittance subsystem A whose mechanical elements can be represented by a target impedance model.

For a single degree-of-freedom (DOF) impedance-controlled robot manipulator shown in Fig. 4, which encompasses many of the basic important factors that affect the stability and performance of robot interaction, the dynamics can be represented as

$$F_i(t) = M_t \dot{v}_i(t) + B_t v_i(t) + K_t x_i(t) + F_d^h(t) \quad (7)$$

$$v_i(t) = v_d(t) \quad (8)$$

where the position-controlled robot G_p is assumed to be effectively cancelled out, x_i is an actual position of the robot, and $v_i = \dot{x}_i$.

Substituting (7) into (4),

$$E_A[n] = V_A[n] - V_A[0] + \int_0^{nT} B_t v_d^2(t)\,dt \quad (9)$$

where $V_A[n] = M_t v_d^2[n]/2 + K_t x_d^2[n]/2$. The third term in (9) means the dissipated energy of the admittance subsystem. This dissipation capability of the admittance subsystem is important for EBA because *EBA can passify the generated energy utilizing this dissipation capability of the admittance subsystem*.

Thus, using (9) and noting that $V_{\mathrm{ASCE}} = V_A + V_{\mathrm{SCE}}$, the passivity condition (6) can be rewritten as

$$\int_0^{nT} B_t v_d^2(t)\,dt + \sum_{k=0}^{n-1} F_d[k]\,\Delta x_d[k+1] \geq V_{\mathrm{CE}}[n] - V_{\mathrm{CE}}[0] \quad (10)$$

where $V_{\mathrm{SCE}} = V_{\mathrm{CE}}$ since there is no storage element in the sample-and-hold subsystem. Using Cauchy–Schwarz inequality $(\int_0^{nT} B_t v_d^2(t)\,dt \geq \frac{B_t}{T}\sum_{\mathrm{k}=0}^{\mathrm{n}-1} \Delta x_d^2[k+1])$, (10) can further be rewritten as

$$\frac{B_t}{T}\sum_{k=0}^{n-1} \Delta x_d^2[k+1] + \left\{\sum_{k=0}^{n-1} (F_d[k] - F_d[k+1])\,\Delta x_d[k+1]\right\} + \sum_{k=0}^{n-1} F_d[k+1]\,\Delta x_d[k+1] \geq V_{\mathrm{CE}}[n] - V_{\mathrm{CE}}[0]. \quad (11)$$

Dividing (11) into two equations, Condition 1 and Condition 2 can be defined as follows:

Condition 1:

$$\frac{B_t}{T}\sum_{k=0}^{n-1} \Delta x_d^2[k+1] + \sum_{k=0}^{n-1} (F_d[k] - F_d[k+1])\,\Delta x_d[k+1] \geq 0. \quad (12)$$

Condition 2:

$$\sum_{k=0}^{n-1} F_d[k+1]\Delta x_d[k+1] \geq V_{\mathrm{CE}}[n] - V_{\mathrm{CE}}[0]. \quad (13)$$

Then, simultaneous satisfaction of Condition 1 and Condition 2 is sufficient condition for (11).

The second passivity condition in (13) is different from the second passivity condition of the haptic EBA in [16] by an additional energy term $V_{\mathrm{CE}}[n]$. This additional term allows us to design a new adaptive EBA that contains online adaptable c_2 parameter, which will be introduced later. Since the haptic EBA in [16] utilizes the dissipation capability of the haptic device to passify the energy generation of the sample-and-hold by simultaneously satisfying Conditions 1 and 2, the proposed adaptive EBA also has to regulate the energy of *CE* subsystem by the controller in Fig. 3(a). For this purpose, the proposed adaptive EBA controls the contact force of the environment using the following difference recursive formula:

Control Law:

$$F_d[n] = F_d[n-1] + \beta[n]\,\Delta x_d[n] \quad (14)$$

where

$$\beta[n] = \frac{F_e[n] - F_d[n-1]}{\Delta x_d[n]} \text{ for } \Delta x_d[n] \neq 0 \quad (15)$$

where $F_e[n]$ is the sampled real contact force from the F/T sensor. Note that if $\Delta x_d[n] = 0$, meaning that the robot is paused, so the control force $F_d[n]$ is kept constant from (14), which is natural in real contact cases. The command force input $F_d[n]$ to the impedance controller should be regulated by the $\beta[n]$ parameter that represents "controllable (or instantaneous) stiffness". To satisfy the passivity conditions in (12) and (13) simultaneously, this parameter may be bounded by the following bounding laws:

Bounding Laws:

$$\text{If } \beta[n] > \beta_{\max}[n], \text{ then } \beta[n] = \beta_{\max}[n].$$

$$\text{If } \beta[n] < \beta_{\min}[n], \text{ then } \beta[n] = \beta_{\min}[n],$$

where $\beta_{\max}[n] = \min(c_1, \gamma_{\max}[n]), \beta_{\min}[n] = \gamma_{\min}[n]$,

$$\gamma_{\max}[n] = c_2 - \frac{F_d[n-1]}{\Delta x_d[n]} + \sqrt{c_2^2 + \left(\frac{F_d[n-1]}{\Delta x_d[n]}\right)^2}. \quad (16)$$

$$\gamma_{\min}[n] = c_2 - \frac{F_d[n-1]}{\Delta x_d[n]} - \sqrt{c_2^2 + \left(\frac{F_d[n-1]}{\Delta x_d[n]}\right)^2} \quad (17)$$

and where c_1 and c_2 are positive parameters satisfying $c_1 \geq c_2$. Detailed derivations and proofs of these control and bounding laws along with ways of determining c_1 and c_2 parameters are explained in the following sections.

D. Determination of Parameters of the Impedance Controller

The design of an impedance controller can follow the procedures of the robust control method [9] for the estimated environmental parameters before designing the adaptive EBA. It is then expected that the robot can be stable against the conditions that were considered in the impedance controller design.

E. Determination of Parameters of the Adaptive EBA

For robust stability and good performance, we suggest procedures for determining the values of c_1 and c_2 that are two main control parameters of the proposed adaptive EBA. These procedures also include formal proofs of the passivity conditions in (12) and (13).

Lemma 1: With the control law in (14), the Condition 1 is satisfied if $\beta[n]$ is bounded such that

$$\beta[n] \leq c_1 \leq B_t/T \text{ for } \forall n \geq 0. \quad (18)$$

Proof: The left-hand side of (12) can be represented by the control law in (14) as

$$\sum_{k=0}^{n-1}\left(\frac{B_t}{T} - \beta[k+1]\right)\Delta x_d^2[k+1] \geq 0. \quad (19)$$

Therefore, if $\frac{B_t}{T} \geq c_1 \geq \beta[k+1]$ for all k, (19) is satisfied. Note that the parameter has a dimension of stiffness. ■

Note first that the parameter c_1 is not related to the environment but is only related to (more specifically, is upper-bounded by $c_1 \leq B_t/T$) the energy dissipation capability of the impedance-controlled industrial robot as shown in Lemma 1. The energy dissipation capability is a function of frictions in the robotic system and is dependent on the robot pose because the proposed passivity condition is formulated in the task space. Because of linear approximation of the impedance-controlled robot dynamics [as in (7)] and because of neglecting unmodeled dynamics (e.g., Coulomb friction, etc) in deriving Lemma 1, the parameter c_1 cannot simply be determined by B_t/T. A practical way of finding this parameter of the robot is that it is gradually increased until sustained oscillations occur or is gradually decreased until no oscillations occur from an arbitrary initial guess, while contacting a very stiff environment that causes instability against nominal impedance-controlled industrial robot. This is because this high stiffness environment is considered as the worst case with respect to the stability to find the maximum c_1 value. This process can then be executed offline over the operation workspace. Thus, it is not determined by a specific environment but is determined by an arbitrary high stiff environment. Therefore, we do not require exact models of the environments to find out c_1 parameter. This offline parameter estimation, therefore, also does not need to consider robot modeling errors.

Inaccurate c_1 parameter estimation may jeopardize the stability or may degrade the performance. If this parameter is overestimated, it may cause instability when the industrial robot interacts with the environments with higher stiffness than the true parameter c_1. In order to avoid this situation, accurately estimated or more conservative c_1 (i.e., underestimated c_1) must be used for stability. Even though the stability is still satisfied with the underestimated c_1, the performance may be deteriorated because the maximum value of the adaptive c_2 is limited by the c_1 value [see (29)]. Nevertheless, some performance such as steady-state desired contact force may be obtained by adjusting the commanded penetration depth as explained in the experimental results.

Note that the c_1 value decided this way can be used for all contacts with different stiffness environments because this parameter is only related to the maximum energy dissipation capability of the plant. Moreover, because of its independence from the environmental stiffness, c_1 must not be estimated for soft environments but be estimated against a very stiff environment for getting the maximum capability. Adaptive c_1 for environments with different stiffness values, therefore, is not necessary too. If the parameter c_1 is estimated against a soft environment, it may give incorrect parameter so that it may cause instability.

Lemma 2: Condition 2 is satisfied if the following inequality holds true:

$$F_d[n]\,\Delta x_d[n] \geq \frac{1}{2c_2}\left(F_d^2[n] - F_d^2[n-1]\right) \quad (20)$$

where c_2 is positive, which has a dimension of stiffness.

Proof: With (20), the left-hand side of (13) in Condition 2 can be rewritten as

$$\sum_{k=0}^{n-1} F_d[k+1]\,\Delta x_d[k+1] \geq \frac{1}{2c_2}\sum_{k=0}^{n-1}\left(F_d^2[k+1] - F_d^2[k]\right)$$
$$= \frac{1}{2c_2}\left(F_d^2[n] - F_d^2[0]\right). \quad (21)$$

Therefore, Condition 2 is satisfied if

$$\frac{1}{2c_2}\left(F_d^2[n] - F_d^2[0]\right) \geq V_{\mathrm{CE}}[n] - V_{\mathrm{CE}}[0]. \quad (22)$$

Assuming $F_d[0] = 0$ and $V_{\mathrm{CE}}[0] = 0$ *where n is set to zero just before contact*, (22) states that if a positive parameter c_2 can be chosen such that

$$c_2 \leq \frac{1}{2}\frac{F_d^2[n]}{V_{\mathrm{CE}}[n]} \quad (23)$$

then, Condition 2 is satisfied. ■

Since the energy $V_{\mathrm{CE}}[n]\ (= \sum_{k=0}^{n} F_e[k]\,x_d[k])$ is measurable by the measured force (F_e) and penetration depth (x_d), a positive parameter c_2 can be adaptively tuned by (23). For example, if the environment is static and a pure spring with stiffness K_e, which is a typical real environment and is also the worst case for stability, then $F_d[n] \leq F_e[n] = K_e x_d[n]$ because the desired maximum command force $F_d[n]$ is the environmental contact force $F_e[n]$. In other words, when unbounded, $F_d[n] = F_e[n]$from (14) and (15). In this case, $V_{\rm CE}[n] = \frac{1}{2} K_e x_d^2[n]$ and then from (23), c_2 is obtained as

$$c_2 \leq K_e. \tag{24}$$

If the parameter c_2 can be set to the upper limit K_e, which is desired for contact performance, then, this *c_2 parameter can be considered as a stiffness estimate for the environment*.

The inequality condition in (20) can be made true by bounding the computed $\beta[n]$ as follows.

Lemma 3: With the control law in (14), Condition 2 in (13) is satisfied if $\beta[n]$ is bounded as

$$\gamma_{\max}[n] \geq \beta[n] \geq \gamma_{\min}[n]. \tag{25}$$

Proof: Inserting the control law in (14) into (20) gives

$$\{F_d[n-1] + \beta[n]\Delta x_d[n]\}\Delta x_d[n]$$
$$\geq \frac{1}{2c_2}\left(\{F_d[n-1] + \beta[n]\Delta x_d[n]\}^2 - F_d^2[n-1]\right). \tag{26}$$

Rearranging (26) into the second-order inequality equation for $\beta[n]$ as

$$\Delta x_d^2[n]\beta^2[n] + 2\left(F_d[n-1]\Delta x_d[n] - c_2\Delta x_d^2[n]\right)\beta[n]$$
$$-2c_2 F_d[n-1]\Delta x_d[n] \leq 0.$$

The above inequality equation always has a solution as

$$(\beta[n] - \gamma_{\max}[n])(\beta[n] - \gamma_{\min}[n]) \leq 0 \tag{27}$$

where $\gamma_{\max}[n]$ and $\gamma_{\min}[n]$ are defined in (16) and (17), respectively, which proves (25). ■

Finally, the passivity condition (6) is satisfied in a feasible region of $\beta[n]$ satisfying both conditions in (18) and (25) as follows.

Theorem 1: Conditions 1 and 2 can be satisfied simultaneously if there exists a region of $\beta[n]$ such that

$$\min(c_1, \gamma_{\max}[n]) \geq \beta[n] \geq \gamma_{\min}[n]$$
$$\text{with a choice of } c_2 \text{ such that } c_1 \geq c_2. \tag{28}$$

Proof: Noting that a positive parameter c_2 is always between $\gamma_{\max}[n]$ and $\gamma_{\min}[n]$ from (16) and (17) in the bounding laws because $\gamma_{\max}[n] - c_2 \geq 0$ and $c_2 - \gamma_{\min}[n] \geq 0$ for all nonnegative n, there is no intersection of $\beta[n]$ satisfying both Conditions 1 and 2 if c_1 is smaller than $\gamma_{\min}[n]$ as seen in the c_1 location ① in Fig. 5. This situation can, however, be overcome if c_2 is chosen such that $c_1 \geq c_2$ (and thus, $c_1 \geq c_2 \geq \gamma_{\min}[n]$), which guarantees $c_1 \geq \gamma_{\min}[n]$. Then, there is always a feasible region of $\beta[n]$ satisfying both Conditions 1 and 2 as seen in the c_1 location ② in Fig. 5. ■

Since c_2 must be smaller than c_1 from Theorem 1 as well as $c_2 \leq K_e$ from (24) for a static and pure spring environment, c_2 can be selected from the following condition:

$$c_2 \leq \min(K_e, c_1). \tag{29}$$

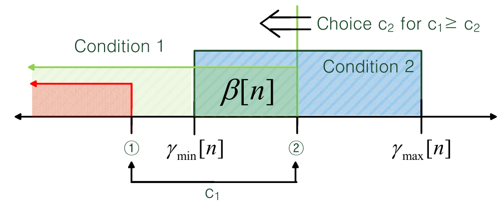


Fig. 5. Feasible region satisfying both $\beta[n]$ in (18) and in (25).

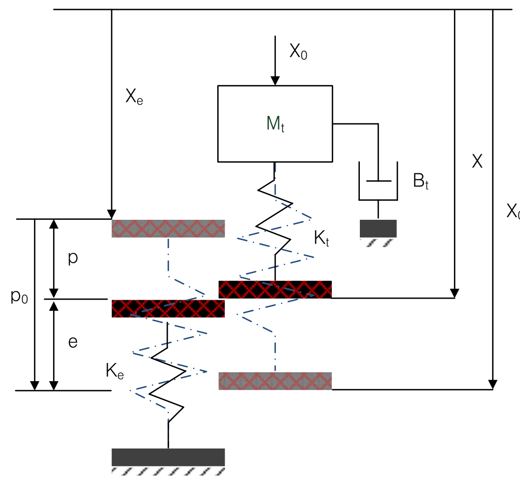


Fig. 6. Unilateral contact process of impedance-controlled robot with a pure stiffness environment that is considered as the worst case environment [7].

Note that when identifying the parameter c_1 in the offline estimation process during the robot is contacting a very stiff wall [therefore, $c_1 < K_e$ and $c_2 = c_1$ *by* (29)], the value of c_2 needs to be set to c_1 as required by the passivity condition in (29). Later on, in the online force control process, c_2 is going to be adaptively adjusted to the estimated environmental stiffness K_e as follows: Depending on the magnitude of the environmental stiffness (K_e) and the dissipation capability (c_1) of the impedance-controlled robot, we can consider the following two cases:

1) *Case I:* If $c_1 \geq K_e$ (and thus, $c_1 \geq K_e \geq c_2$ from (24)), then c_2 can be set to the real environmental stiffness K_e. This means that the desired contact force can fully be maintained because the energy dissipation capability (represented by c_1) of the impedance-controlled robot is beyond the required (K_e).
2) *Case II:* If $c_1 \leq K_e$ (and thus, $c_2 \leq c_1 \leq K_e$ from Theorem 1), then, c_2 must be set to c_1. This means that the desired contact force is limited by the limited stiffness value c_1 due to the limited energy dissipation capability of the impedance-controlled robot.

Steady-state contact force responses for the aforementioned two cases will further be explained in the following performance analyses.

F. Performance Analysis of the Adaptive EBA

To investigate the performance of the proposed adaptive EBA, consider Fig. 6 showing a single DOF contact process of impedance-controlled robot with a passive environment with stiffness K_e. Before analyzing performance of the proposed adaptive EBA, we first investigate the performance of the PBIC to compare the haptic EBA and the adaptive EBA.

From Fig. 6, the steady-state contact force of the PBIC can be calculated as

$$F_e^* = K_e p^* = K_e \left(p_0 - e^*\right) = K_e \left(p_0 - \frac{F_e^*}{K_t}\right) \tag{30}$$

where a superscript $*$ symbol means steady state, p_0 is the nominal penetration depth, p is the actual penetration into the environment, and e^* is the steady-state position correction of the impedance controller. This nominal path modification e^* can be computed by $\lim_{s\to 0} G_f\left(s\right) F_e^*$, where $\lim_{s\to 0} G_f\left(s\right) = 1/K_t$ from (1) and Fig. 1. The steady-state contact force F_e^* in (30) can then be computed as

$$F_e^* = \frac{K_t K_e}{K_t + K_e} p_0. \tag{31}$$

Note that the combined spring $\left(\frac{K_t K_e}{K_t + K_e}\right)$ is a resultant spring from a series two springs K_e and K_t.

On the other hand, the steady-state contact force of the proposed adaptive EBA with PBIC can be considered for the two cases depending on the environmental stiffness:

1) *Case I:* $c_1 \geq K_e \geq c_2$

This case usually corresponds to a soft environment case so that c_2 can be set to the real environmental stiffness K_e, and $\beta\left[n\right]$ is mostly unbounded because the energy dissipation capability (represented by c_1) of the impedance-controlled robot is larger than the required $\left(K_e\right)$. In this case, the steady-state force $\left(F_d^*\right)$ of the adaptive EBA can be equal to the environment contact force $\left(F_e^*\right)$ from inserting (15) into (14) as

$$F_d^* = F_e^*. \tag{32}$$

This means that the proposed adaptive EBA can show the same performance as that of the original PBIC.

However, in the case of the haptic EBA where a non-adaptive constant c_2 was used, only one stiffness value of the environment can be considered for various environmental stiffness values. As a result, original behavior of the impedance control cannot be preserved in the haptic EBA interacting with other stiffness environments, although contact stability is guaranteed. On the contrary, the parameter c_2 of the proposed adaptive EBA can be adapted to various environmental stiffness values. Therefore, the impedance control performance within normal operations can be preserved.

2) *Case II:* $c_2 \leq c_1 \leq K_e$

This case usually corresponds to a stiff environment case in which $\beta\left[n\right]$ is mostly upper bounded by c_1 to prevent the system from instability. The steady-state output force of the adaptive EBA in this case can be explained by the block diagram in Fig. 7 showing the bounded EBA force F_d.

Assuming $X_e = 0$ without loss of generality, the transfer function from X_0 to F_d can be approximated as

$$\frac{F_d\left(s\right)}{X_0\left(s\right)} = \frac{G_p\left(s\right) c_1}{1 + G_p\left(s\right) G_f\left(s\right) c_1} \approx \frac{G_p\left(s\right) c_1}{1 + c_1/Z_t} \tag{33}$$

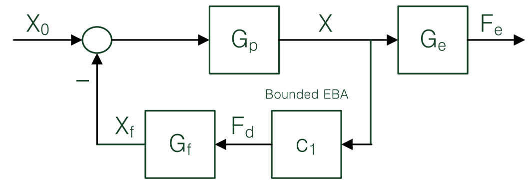


Fig. 7. Bounded EBA block diagram.

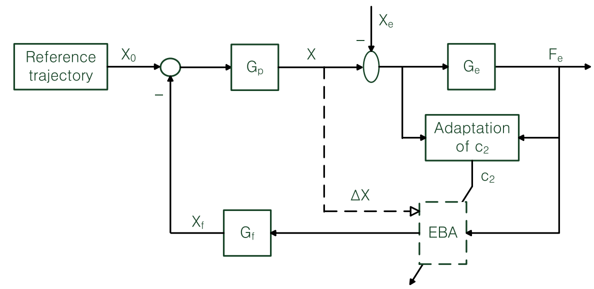


Fig. 8. Total proposed control block diagram.

where Z_t means the target impedance from (1). For step input X_0, steady-state force $\left(F_d^*\right)$ of the adaptive EBA can be approximated as

$$F_d^* \approx \lim_{s\to 0} \frac{G_p\left(s\right) c_1}{1 + c_1/Z_t} x_0 = \frac{c_1}{1 + c_1/K_t} p_0 = \frac{K_t c_1}{K_t + c_1} p_0 \tag{34}$$

where $G_p\left(0\right) = 1$ and $X_0 = p_0$ due to $X_e = 0$. In this case, the steady-state position correction e^* $\left(= X_f^*\right)$ in Fig. 7 from the impedance controller can be derived as

$$e^* = F_d^*/K_t. \tag{35}$$

Inserting F_d^* in (34) into (35), the steady-state contact force F_e^* for the bounded EBA case can be derived as

$$\begin{aligned} F_e^* &= K_e p^* = K_e\left(p_0 - e^*\right) = K_e\left(p_0 - \frac{F_d^*}{K_t}\right) \\ &\approx K_e\left(p_0 - \frac{c_1}{K_t + c_1} p_0\right). \end{aligned} \tag{36}$$

Finally, the steady-state contact force is obtained as

$$\therefore\ F_e^* \approx \frac{K_t K_e}{K_t + c_1} p_0. \tag{37}$$

Comparing the steady-state contact force in (31), the denominator of (37) is smaller than the denominator of (31). Therefore, (37) generates a larger contact force than the desired one. This performance degradation can, however, be compensated easily by adjusting nominal penetration depth p_0. This is possible because the adaptive EBA can provide absolute contact stability for any passive environments so that a desired contact force can be obtained by regulating reference trajectory, while keeping contact when the environment stiffness will turn out to be larger than the designed value. This strategy can be called *prestability and postperformance*.

Fig. 8 shows the entire block diagram of the proposed adaptive EBA, where the block with adaptation of c_2 estimates the environmental stiffness K_e using the F/T sensor. Many estimator techniques may be used but the following simple Hook's law is used in the current investigation:

$$K_{\text{estimate}} \approx \frac{F_e - F_{\text{th}}}{x - x_e} \tag{38}$$

where F_{th} is a threshold value both to check contact status and to identify the contact location X_e that can be obtained when the contact force exceeds F_{th}. The parameter c_2 is then updated by (29) with the estimated stiffness K_{estimate} in (38).

In the existing adaptive impedance control methods [10], [15], the impedance parameters are adapted based on the estimated environmental parameters or contact force signals during the operation. In these methods, therefore, time lags from online estimation and adaptation may cause stability problem. Furthermore, inaccurate estimation or adaptation may have bad influence on the stability. In the proposed adaptive EBA, however, the time delay and accuracy associated with environmental stiffness estimation do not cause any stability problem because the passivity condition that is proved in Section III-E has nothing to do with any time delays or inaccuracy from online adaptation of c_2 as long as c_2 is upper bounded by c_1. This is why the EBA is so robust against the uncertainties and time delays. The stability (passivity) of the proposed adaptive EBA may be affected only by the accuracy of the parameter c_1 but this is not a severe problem in practice because this energy dissipation capability can be experimentally determined offline as discussed in Section III-E.

IV. Experiments

A. Experimental Setup

In this section, experimental procedures and results are presented to demonstrate the performance of the proposed adaptive EBA. Typical contact tests that are similar in [9] are performed for the environments with various stiffness values (nominal, stiffer, and softer). The experimental system is composed of the six-axis DENSO robot [21], robot's own position controller, personal computer, F/T sensor (six-DOF ATI mini45 sensor [22]), and a real environment as shown in Fig. 9(a). In the experiments as shown in Fig. 9(b), the robot is contacting in the vertical direction to a thin cantilevered plate that can simulate various contact stiffness values (soft 50 kN/m at the free end; nominal 100 kN/m at the midpoint; stiff 180 kN/m at the fixed end). The impedance controller is designed for the following control conditions: $p_0 = 0.002$ (m), $F_{\text{desired}} = 10$ (N), $K_e = 100$ (kN/m), where p_0 is the desired penetration depth that is to be commanded to the impedance-controlled robot, F_{desired} is the desired contact force, and K_e is the assumed nominal stiffness of the environment.

Since an estimated position-controlled robot transfer function $\tilde{G}_p$ of the given industrial robot is necessary to implement the improved impedance control law in (1) for the SISO system, system identification for the Denso robot is carried out first. The designed impedance controller about the vertical z-axis for 1 DOF is obtained as

$$G_f(s) = \frac{s^2 + 19.75s + 199}{199} \frac{1}{0.106s^2 + 7.828s + 5.263} \tag{39}$$

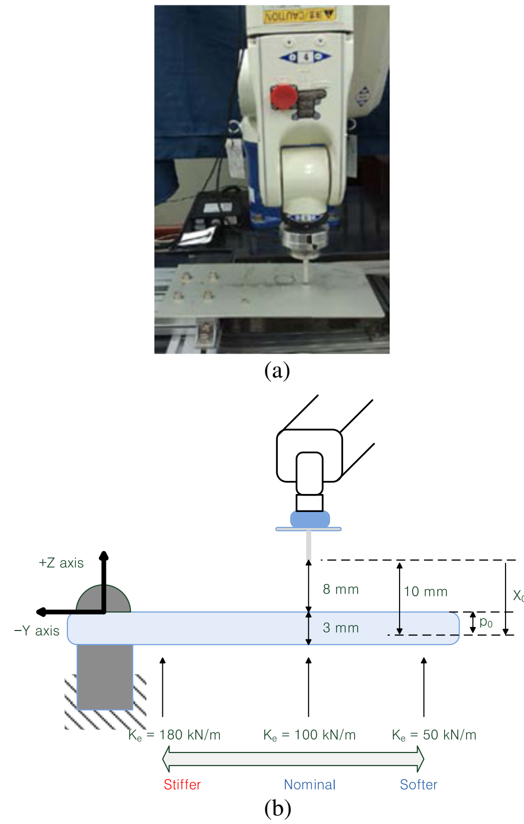


Fig. 9. Experimental setup.

where the desired damping ratio $\zeta_t \fallingdotseq 5.25$ in the impedance controller and the estimated position-controlled robot transfer function $\tilde{G}_p(s) = \frac{199}{s^2+19.75s+199}$. Note that the outer loop that includes the impedance controller and the proposed EBA runs at approximately 50 Hz.

B. Unilateral Contact Tests With Three Control Algorithms

Three control laws (single PBIC, haptic EBA, and proposed adaptive EBA) are compared to show the effectiveness of the proposed adaptive EBA. In the experiments, parameter c_1 is determined ($c_1 = 120$ kN/m) by using the method explained in Section III-E. Parameter c_2 is set equal to c_1 before experiments begin in the case of the haptic EBA but is varied (adapted) online in case of the proposed adaptive EBA.

Fig. 10 shows typically measured interaction forces showing the performance of the single PBIC, the haptic EBA, and the proposed adaptive EBA for nominal, soft, and stiff environmental stiffness values, respectively. Fig. 10(a) shows contact forces with the PBIC alone and the PBIC with the proposed adaptive EBA for the nominal stiffness case (100 kN/m at the midpoint contact). As explained theoretically, both the controllers show very similar results because c_2 values are mostly smaller than c_1 parameter as shown in the bottom plot of Fig. 10(a). In addition, the proposed approach provides approximate desired contact force (10 N) despite the simple environment estimator (38). The proposed adaptive EBA with PBIC, therefore, can show similar contact behavior with that of the single PBIC for the nominal stiffness that was used in the design of the impedance controller.

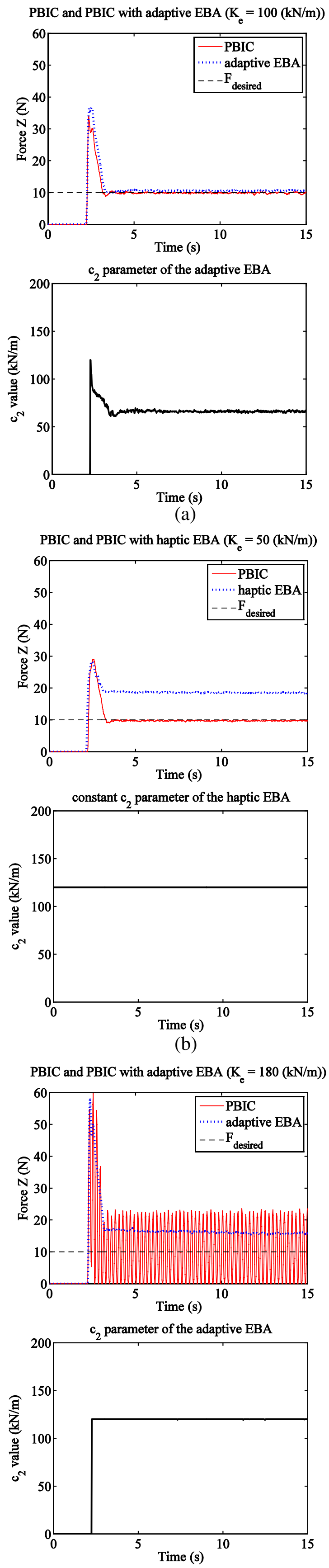


Fig. 10. Contact experiment results with three different stiffness cases: (a) nominal, (b) soft, and (c) stiff.

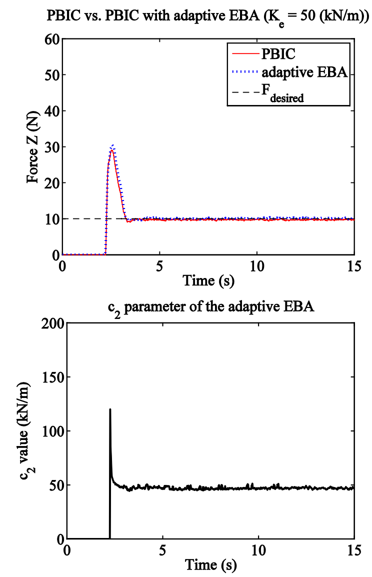


Fig. 11. Effect of adapting c_2 parameters for the soft case.

If the actual environmental stiffness is smaller than the assumed stiffness in the design phase (say, 50% less; 50 kN/m at the free end contact), then, the interaction system is stable because the impedance controller is designed for a stiffer environment. However, the PBIC with haptic EBA shows a large force error between the desired and the actual contact forces [see the dotted line in the top plot of Fig. 10(b)] because c_2 is set to a large constant value c_1 (120 kN/m) in the haptic EBA case. To get the desired contact force, the c_2 value should be adjusted online to the environmental stiffness (50 kN/m) as proposed in Case I of the performance analysis in the Section III-F. Fig. 11 shows this adaptation result.

If the impedance controller that is designed for the nominal stiffness is applied to an environment that turns out to be a lot stiffer (say 80% stiffer than the nominal case: 180 kN/m at the fixed end contact), then unstable contact may be caused as seen in the top plot of Fig. 10(c). The proposed adaptive EBA, however, stabilized the contact, as shown in the top plot with dotted line of Fig. 10(c), without redesign of the impedance controller for the actual environment. However, the measured contact force [about 15 N at the steady-state from (37)] is greater than the desired contact force (10 N) since parameter c_2 is bounded by c_1 in this case [see the bottom plot of Fig. 10(c)] as discussed in the Case II of the performance analysis in the Section III-F.

In order to obtain the desired contact force in the case of Fig. 10(c), the reference position X_0 may be reduced while maintaining stable contact that is guaranteed by the EBA as shown in Fig. 12. The strategy for reference trajectory modification is simple and intuitive: First, monitor the contact force via a *F/T* sensor to check a force error. For the positive force error, reduce the penetration depth. Otherwise, increase the penetration depth. If the sign of the force error is changed, then, reduce the position increment by one half. These procedures are iterated until asymptotically achieving the desired contact force. Therefore, an offline estimation of the environmental parameters is not needed in the proposed control strategy.

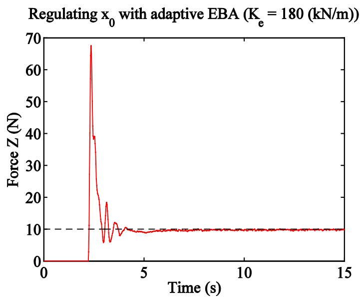


Fig. 12. Effect of regulating nominal penetration depth X_0.

V. Conclusion

This paper presented a passivity-based adaptive theory that extended the haptic EBA in order to improve contact performance of the PBIC industrial robot that may contact with various stiffness environments. Based on a network model of the impedance-controlled robot and the passivity theory, control and bounding laws with adaptive parameter c_2 were proposed. Moreover, methods of determining two important parameters (c_1 and c_2) were presented with the associated theory. For example, the stability parameter c_1 can be decided by an offline method, i.e., by the contact experiment with an arbitrarily high stiffness environment to find maximum c_1. However, as the tradeoff between guaranteed stability and degraded performance, the parameter c_1 also affects the performance because c_2 must be bounded by c_1. For achieving the desired contact force, while contacting softer environment than the nominal stiffness, the parameter c_2 could be updated by environment estimator online. However, when the environmental stiffness is larger than c_1 parameter, the reference position command is proposed to be reduced to get the desired contact force, while guaranteeing the contact stability by the proposed adaptive EBA.

Simple contact experiments with a commercial industrial robot demonstrated the effectiveness of the proposed ideas. This showed applicability of the proposed idea in an actual system. This is possible mainly because the parameter c_1 (i.e., as an inherent robot parameter) is estimated offline against an arbitrary high stiff environment and because the parameter c_2 is adapted online without violating contact stability as long as c_2 is restricted less than c_1. An important benefit of the proposed adaptive EBA, therefore, is that the proposed approach can provide absolutely robust contact stability against any uncertain environments. Moreover, adaptive actions can preserve the original performance of the single PBIC for nominal and soft stiffness values.

In future works, more comprehensive experiments are needed with the industrial robot against many real cases such as peg-in-hole insertion, digging a ground, etc., for showing full reliability of the proposed method. In addition, other various applications including the teleoperation that contains effect of time-varying communication delays are also planned: for example, biped robots contacting with very uncertain grounds and wearable robots contacting with human beings that may change significantly contact impedances. Furthermore, more sophisticated environment estimator for more complex structures will be studied for the better performance of the proposed adaptive EBA.

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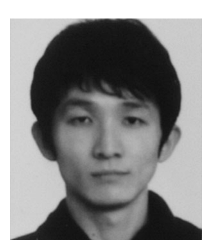

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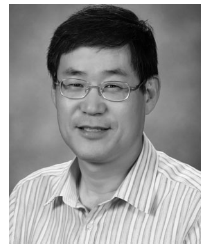

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